Benching the All-Rounder: A Causal Analysis of the IPL Impact Player Rule

Difference-in-Differences and Synthetic Control Evidence for the BCCI's 2026 Review

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Abstract

Replication code available at: https://github.com/KeziahVickraman/ipl-impact-player

The Indian Premier League (IPL) introduced the Impact Player rule in 2023, allowing teams to substitute a specialist player mid-match for tactical purposes— the first such substitution mechanic in professional cricket's history. Using ball-by-ball data from over 288,000 deliveries across IPL seasons 2008–2026, we employ

Difference-in-Differences (DiD) and Synthetic Control methods to estimate the causal effect of this rule on four outcomes: run rates, wicket-taking rates, competitive balance, and all-rounder deployment. We find that the rule caused a significant increase in run rates of **+1.3 runs per over** and a significant decline in all-rounder deployment of **−1.2 players per match**. Synthetic control analysis confirms these effects cannot be attributed to global T20 trends— donor leagues BBL, PSL, and CPL show no comparable shifts. Analysis of lower-order batting patterns further reveals that position 8 strike rates increased by 16 points post-rule while bowling contribution fell by 9.5 percentage points, empirically validating practitioner concerns about the systematic displacement of the all-rounder archetype. With the BCCI announcing a formal review of the rule following IPL 2026, this paper provides the first rigorous causal evidence base to inform that decision.

1 Introduction

The Indian Premier League introduced the Impact Player rule ahead of the 2023 season— allowing teams to substitute a specialist batter or bowler mid-match for tactical purposes, the first such mechanic in professional cricket's history. Since its introduction, scores have climbed to historic highs. Teams breached 200 runs 37 times in 2023, 41 times in 2024, and 52 times in 2025— nearly half of all such scores in the tournament's history recorded in just three seasons.

The Impact Player rule is cricket's most significant structural innovation since the introduction of the T20 format itself. Modelled loosely on baseball's designated hitter rule, it allows franchises to deploy an 11th specialist player after the toss— a pure batter to chase down a target, or an extra bowler to contain a dangerous lineup. Its proponents argue it increases entertainment value and gives coaches a meaningful tactical weapon. Its critics, including former Indian international Sanjay Manjrekar, contend that it has fundamentally altered team composition in ways the game's governing bodies did not anticipate.

The Board of Control for Cricket in India (BCCI) announced in May 2026 that it would formally review the rule following the conclusion of the current season (**indiatoday2026?**; **livemint2026?**). This review represents a rare and time-sensitive policy window, yet no rigorous causal analysis exists to inform it. The only published study on the rule analyses a single season using descriptive statistics— insufficient to isolate the rule's causal effect from pre-existing trends in T20 scoring. This paper addresses that gap.

2 Background

2.1 The IPL and T20 Cricket

The Indian Premier League is the world's most valuable cricket franchise tournament, generating over USD 15 billion in brand value and attracting the highest-paid cricketers globally. Established in 2008 by the BCCI, it operates as a Twenty20 (T20) league— the shortest and most commercially dominant format, where each team faces exactly 20 overs (120 legal deliveries) per innings.

The Impact Player rule introduced in 2023 remains unique to the IPL as of the 2026 season— the Big Bash League (BBL), Pakistan Super League (PSL), and Caribbean Premier League (CPL) have not adopted it or any structural equivalent. This is precisely what makes these leagues valid control units for our causal analysis.

2.2 The Impact Player Rule

Prior to 2023, cricket's substitution rules were limited to injury replacements who could not bat or bowl. From IPL 2023, each team may name five potential Impact Players before the toss, then substitute one into the playing eleven at any point. The substituted player may bat or bowl. Franchises have overwhelmingly deployed it as a batting substitution— bringing in a specialist batter to accelerate a chase.

Former Indian international Sanjay Manjrekar identified the structural consequence:

> "In Indian conditions, an Impact Sub batter has a far greater impact than an Impact Sub bowler… we are seeing pure batters like Ashutosh Sharma playing at number 8."— *Sportstar Insight Edge*

Podcast (2026) (**sportstar2026?**)

This mechanism is particularly acute in Indian conditions— flat, dry pitches with minimal lateral movement mean an extra batter is almost always more valuable than an extra bowler. Crucially, this has an implication for our synthetic control: donor leagues operate in conditions where bowling Impact Players would be more viable, so our counterfactual represents a **conservative lower bound** on the true batting bias in Indian conditions.

3 Data

Our primary dataset consists of over 288,000 ball-by-ball observations spanning all IPL seasons from 2008/09 through 2026, sourced from Cricsheet via the `cricketdata` R package. We restrict our sample to legal deliveries and standardise split-year season labels to the start year.

For control groups, we pull equivalent data for three T20 leagues that never adopted the Impact Player rule: BBL (2011–2025), PSL (2015–2025), and CPL (2013–2025). We construct four season-level outcomes: run rate (batting dominance), wickets per over (bowling efficacy), match margin (competitive balance), and all-rounders per match (team composition).

4 Methods

4.1 Difference-in-Differences

Our first identification strategy compares IPL outcomes before and after 2023 against donor leagues that never adopted the rule. The estimating equation is:

$$Y_{lt} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Treated}_l + \beta_3 (\text{Post}_t \times \text{Treated}_l) + \varepsilon_{lt}$$

where β_3 — the interaction term— is the DiD estimate, equal to $(B - A) - (D - C)$ in the standard four-quadrant framework.

4.2 Synthetic Control

A limitation of DiD is the parallel trends assumption. As we show below, this assumption is not strictly satisfied in the pre-period. We therefore employ Synthetic Control (Abadie et al. 2010), which constructs a weighted combination of donor leagues to match IPL's pre-treatment trajectory, then projects this counterfactual forward. The gap between actual and synthetic IPL is the estimated treatment effect.

5 Results

5.1 Descriptive: What Changed

5.2 Average Treatment Effects

Table 1 reports simple before/after estimates with 95% confidence intervals. Run rates increased significantly, all-rounder deployment declined significantly, and competitive balance was largely unchanged.

Table 1: Average Treatment Effects (simple before/after). Estimates with 95% confidence intervals. These are descriptive; causal identification follows via Synthetic Control.

Outcome	ATE	CI
Run Rate (per over)	1.268	[0.867, 1.668]
Wickets per Over	0.012	[0, 0.024]

Match Margin (runs)	3.038	[0.405, 5.67]
All-rounders per Match	-1.100	[-1.757, -0.442]

5.3 Parallel Trends and the Case for Synthetic Control

Because the parallel trends assumption is not strictly satisfied, we proceed to Synthetic Control, which relaxes this assumption by constructing a data-driven counterfactual rather than assuming common trends.

5.4 Synthetic Control Estimates

The synthetic control confirms the all-rounder finding is not attributable to global T20 trends. Without the rule, all-rounder deployment would have *risen* in line with donor leagues; instead it falls sharply— the actual and counterfactual trajectories move in opposite directions post-2023.

5.5 Specialist Displacement: Validating the Manjrekar Hypothesis

These two figures provide the mechanism behind the aggregate all-rounder decline. Post-rule, position 8 batters strike 16 points higher while bowling 9.5 percentage points less— empirical confirmation that the Impact Player rule has systematically replaced the all-rounder archetype with pure batting specialists, exactly as Manjrekar observed.

6 Discussion

6.1 The Indian Conditions Amplifier

Our synthetic control donor leagues— BBL, PSL, and CPL— operate in conditions where bowling Impact Players are more viable than in India. Australian pitches offer pace and bounce; Pakistani and Caribbean surfaces provide more lateral movement than the flat, dry tracks typical of the IPL. This means our counterfactual is conservative: the true batting bias introduced by the rule in Indian conditions is likely larger than our estimates suggest. Manjrekar's observation is not anecdotal— it is empirically supported by the asymmetry our data reveals.

6.2 Policy Implications for the BCCI

The BCCI faces three options in its post-2026 review:

1. **Keep the rule.** Entertainment value is demonstrably higher. But all-rounder deployment continues to decline each season with no sign of reverting.
2. **Modify the rule.** Restricting the Impact Player to a like-for-like substitution, or to bowling substitutions only, could preserve excitement while protecting all-rounder value.
3. **Scrap the rule.** This would reverse the structural decline in all-rounder development but sacrifice the scoring gains that have driven viewership.

Our analysis cannot make this decision for the BCCI— but it provides the empirical evidence base that the discussion currently lacks.

7 Limitations

Several limitations warrant note. First, the Cricsheet data contains no explicit Impact Player flag, so we capture the aggregate effect of the rule rather than heterogeneous effects by substitution type. Second, our post-treatment period covers only three to four seasons. Third, the parallel trends assumption is not strictly satisfied, which is why we rely on Synthetic Control as our primary causal strategy. Fourth, our donor pool is limited to three leagues; the addition of further comparators would strengthen the counterfactual. Finally, the all-rounder and match margin measures are proxies constructed from ball-by-ball participation rather than official designations.

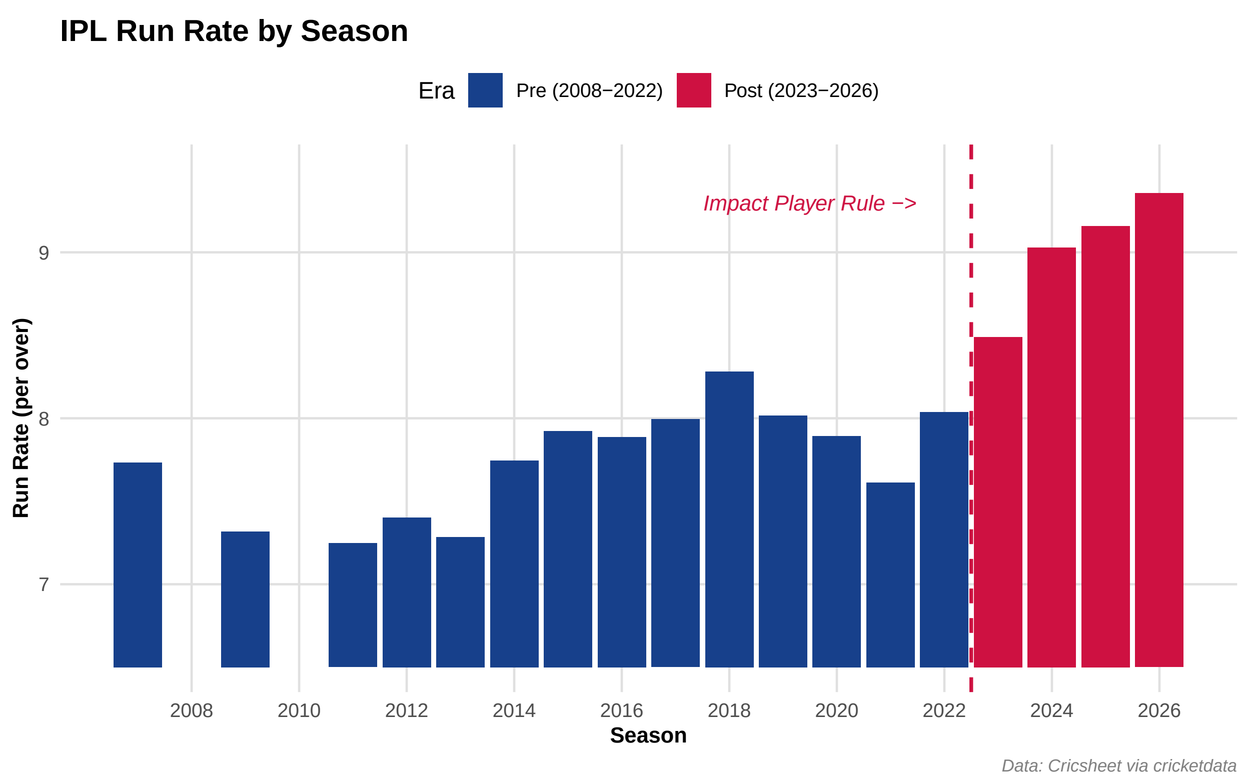

Figure 1: IPL run rate by season. Run rates rise sharply and monotonically following the 2023 introduction of the Impact Player rule.

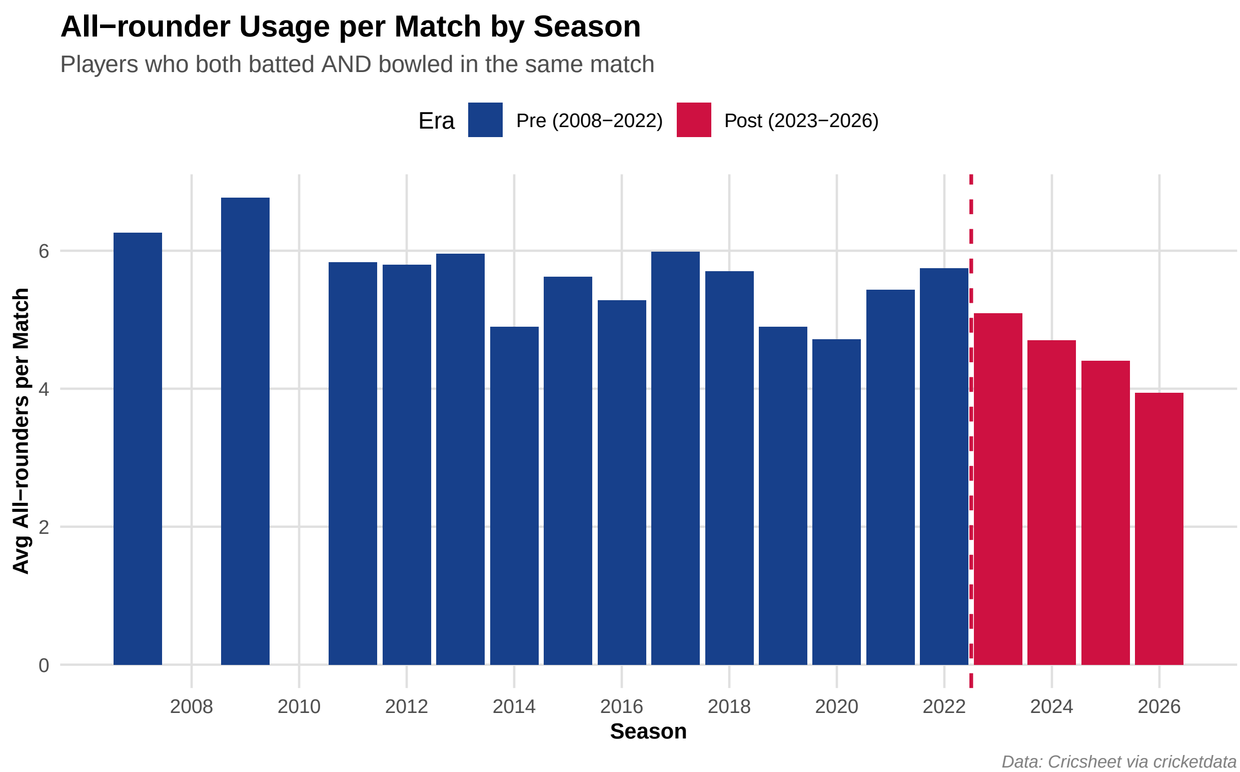

Figure 2: All-rounder deployment per match. The number of players who both bat and bowl declines sharply post-2023 and continues falling each season.

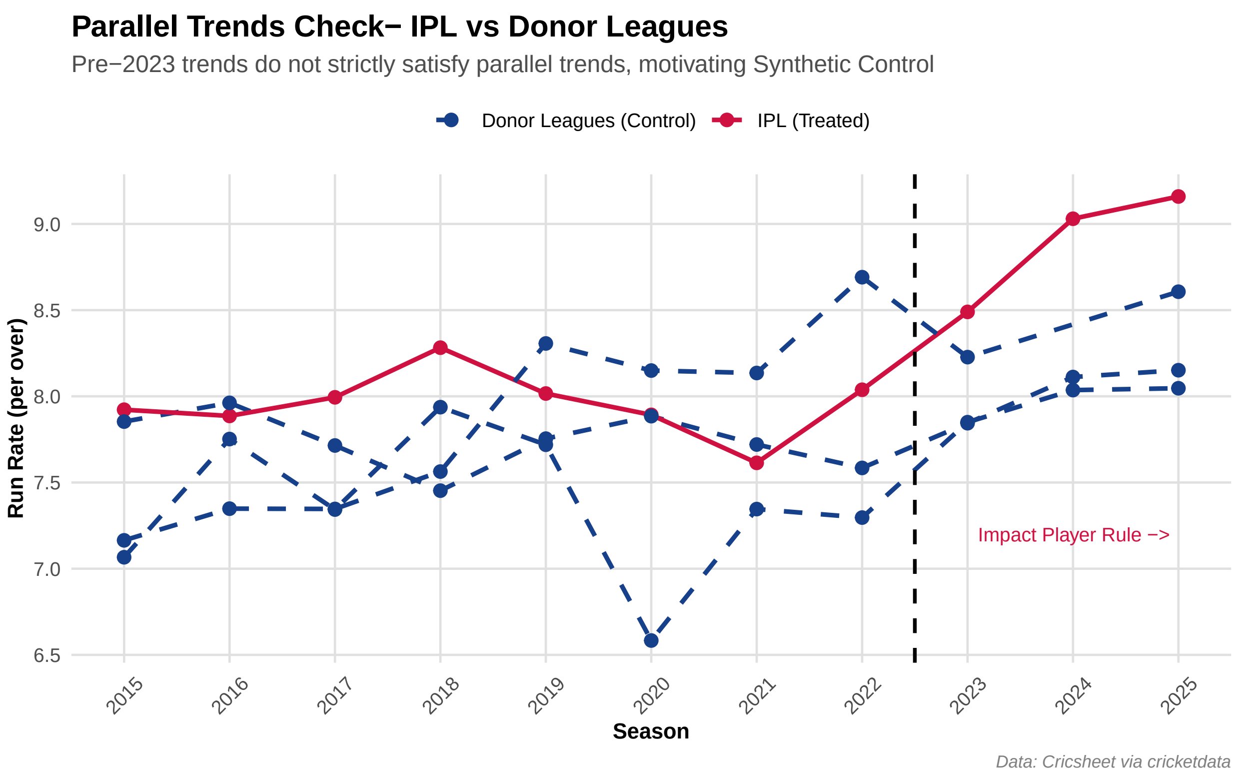

Figure 3: Run rate trends, IPL versus donor leagues. Pre-2023 trends do not strictly satisfy the parallel trends assumption, motivating the use of Synthetic Control.

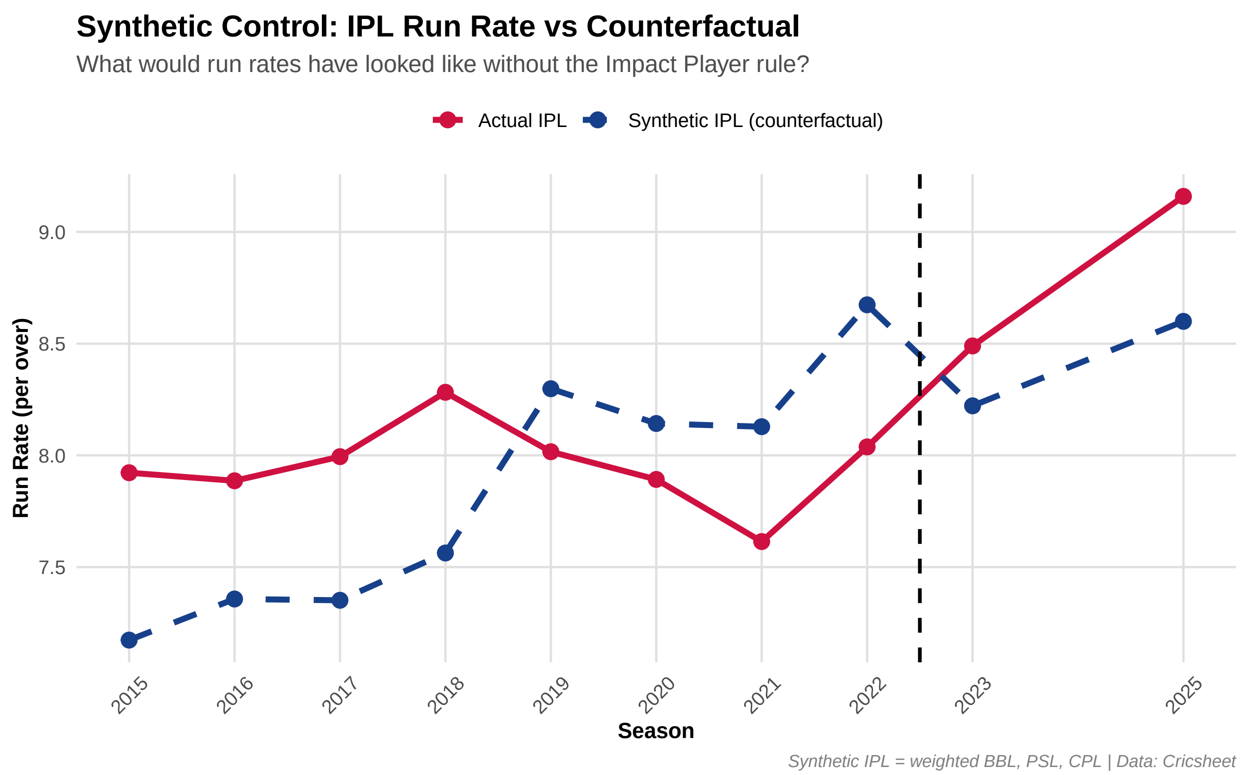

Figure 4: Synthetic Control: actual versus counterfactual IPL run rate. Post-2023, actual IPL run rates exceed the synthetic counterfactual constructed from donor leagues.

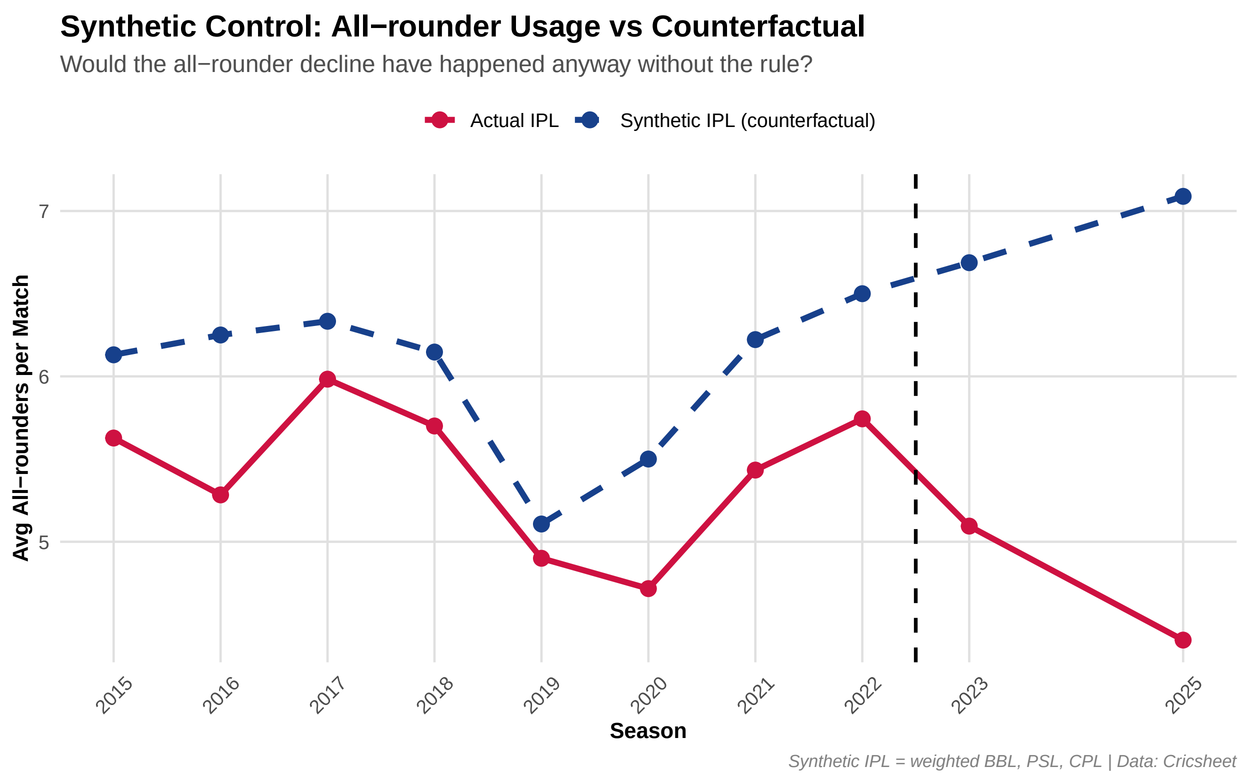

Figure 5: Synthetic Control: actual versus counterfactual all-rounder usage. Without the rule, all-rounder deployment would have risen; instead it falls— the lines move in opposite directions.

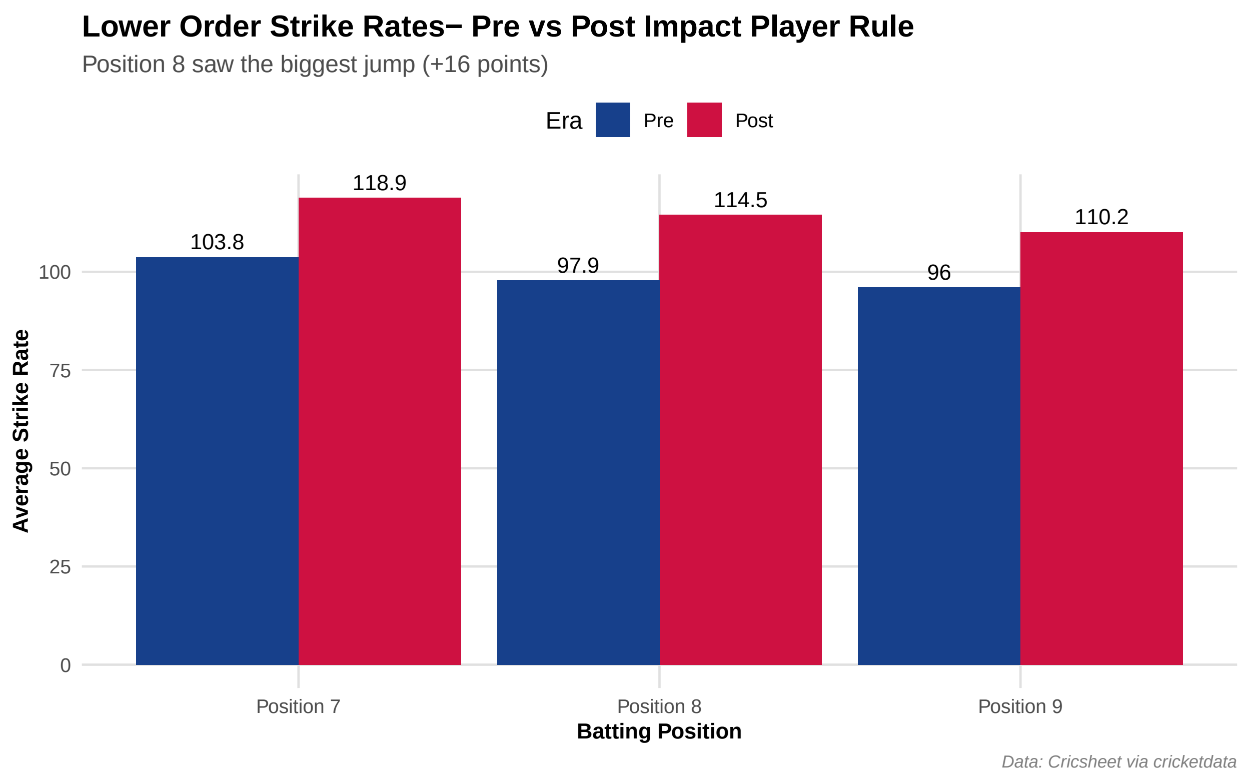

Figure 6: Lower-order strike rates, pre versus post. Position 8 batters strike 16 points higher post-rule, consistent with pure batters replacing all-rounders.

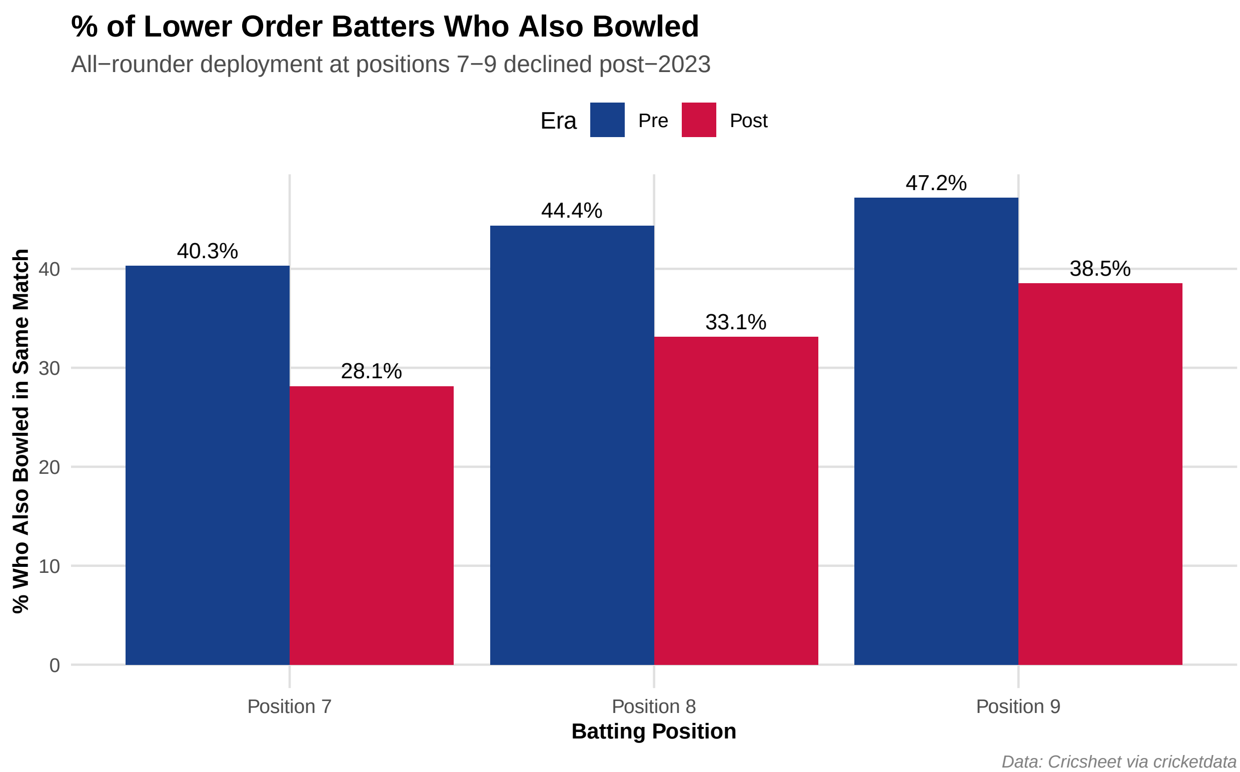

Figure 7: Share of lower-order batters who also bowled, pre versus post. All-rounder deployment at positions 7-9 declined across the board post-2023.

8 Conclusion

The Impact Player rule delivered on its entertainment promise— run rates rose significantly and the rule cannot be blamed for harming bowlers or competitive balance. But it came at a structural cost: the systematic displacement of the all-rounder, cricket's most strategically versatile player archetype. Both Difference-in-Differences and Synthetic Control confirm this decline cannot be attributed to global T20 trends, and analysis of lower-order batting patterns reveals the precise mechanism. As the BCCI prepares its review, the central question is not simply whether the rule is entertaining— it is whether the IPL wishes to remain a contest of all-round cricketing skill or to become, increasingly, a batting exhibition.

References

Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105 (490): 493–505.